GRANT PROPOSAL

KUMBOLA

Kum Bola: kill trash

A ride-hailing platform that coordinates Aboboyaa tricycles for reliable, verified and inclusive waste collection in Greater Kumasi

Submitted by

KoomBei Digital Limited (technology lead), Accra

in partnership with WasteCure Limited (implementation lead), Kumasi

Grant requested: GHS 1,725,206 (US$149,368) over 36 months

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Proposal at a Glance

Project	KumBola: ride-hailing coordination of Aboboyaa tricycles for household, business and institutional waste collection
Applicants	KoomBei Digital Limited, Accra (technology lead) WasteCure Limited, Kumasi (implementation and operations lead, founded 2021)
Proposed partners	Kumasi Metropolitan Assembly and the Suame, Kwabre East and Oforikrom assemblies (MoUs); Sanitation Tricycle Riders Association; a KNUST research team (independent evaluation)
Location	Greater Kumasi Metropolitan Area, Ashanti Region, Ghana. Pilot in Suame, Kwabre East and Oforikrom (about 179,508 households), then Asokwa, Old Tafo and Kwadaso
Duration	A 6-month pre-enrolment research phase (including a tested app prototype), followed by the 36-month project: 5-month inception and build, 7-month pilot, 24 months of scale-up and consolidation
Problem	Greater Kumasi generates about 2,000 tonnes of waste a day. Collection is unreliable, disposal is unverified, the Oti landfill is near capacity, and the tricycle operators who already carry most household waste work informally, unprotected and unregistered
Solution	A web, USSD and mobile money platform that dispatches vetted Aboboyaa operators the way Uber and Bolt dispatch drivers, pays them only after GPS and photo proof that each load reached an approved disposal site, and routes sorted recyclables to WasteCure
Results by month 36	20,000 household accounts (about 72,000 people), 600 businesses, 3 institutional contracts, 127 driver livelihoods, 26,195 tonnes collected with verified disposal, 1,478 tonnes of recyclables recovered, about 2,619 tCO2e avoided
Driver income	GHS 11,584,072 paid to operators over 36 months (6.7 cedis of driver income per cedi of grant). Typical net earnings GHS 2,296 to GHS 3,296 a month, 4 to 6 times the national minimum wage
Financial sustainability	Operating surplus from month 27; Year 3 operating margin 18%; month-36 run-rate surplus GHS 70,405 a month. No operating grant needed after Year 2
Total project cost	GHS 2,025,206 (US$175,343)
Grant requested	GHS 1,725,206 (US$149,368), of which GHS 1,042,397 in Year 1, GHS 627,169 in Year 2 and GHS 55,640 in Year 3
Co-financing	GHS 300,000 in kind (15%): KoomBei software development and WasteCure's anchor fleet and aggregation site. Platform revenue of GHS 2,855,115 is reinvested over the same period
Value for money	Grant cost of GHS 66 per tonne collected over 36 months (GHS 23 over five years), GHS 86 per household served, against an informal collection operating cost of GHS 432 per tonne in Kumasi

1. Executive Summary

Greater Kumasi produces about **2,000 tonnes of solid waste every day** (Kumasi Metropolitan Assembly, as reported by GBC, 2026). Its main disposal site, the Oti landfill, opened in 2004 with a 15-year design life and now receives waste from about 13 municipalities. In January 2026 the Mayor warned that it would be full within 14 to 16 months unless €6 million is found for new cells (MyJoyOnline, 2026a). In June 2026, broken compactors left tricycle waste collectors stranded at the site for four days with full loads, and they threatened to dump at unauthorised locations

(MyJoyOnline, 2026b). The waste that never reaches an approved site ends up burned, buried or in drains. Choked drains are now officially cited as a flooding driver, and Ashanti recorded cholera cases, including one in Kumasi, in the 2024/25 outbreak (3News, 2025; Citinewsroom, 2026a).

The workforce that actually moves Kumasi's household waste is the **Aboboyaa**, the motorised cargo tricycle. In Suame Municipality, **61.5% of households rely on informal collectors**, mostly on tricycles, against 35% on formal companies (Ampong et al., 2024). Aboboyaa collection was introduced by the Kumasi Metropolitan Assembly (KMA) with Zoomlion in 2006, but the sector has stayed informal. Operators negotiate prices on the spot, run loads at 106% of rated capacity, pay GHS 30 a day to KMA plus per-visit tolls, and face health risks: 12.9% of 155 Kumasi tricycle waste operators tested positive for respiratory pathogens (Armoh et al., 2023). Households cannot book them reliably, cannot see where their waste goes, and the city cannot tell a legitimate operator from an illegal dumper.

KumBola ("kum" is Twi for kill and "bola" is Twi for refuse, so the name means kill trash, with a nod to the Koomson family name) applies the model that Uber, Bolt and Yango used to organise Kumasi's taxis to the Aboboyaa. A household or business books a pickup on the web app, by USSD on any phone, on WhatsApp or through a community agent, and pays by mobile money. The platform dispatches the nearest vetted operator. The operator is paid only after the load is **verified at an approved disposal or transfer site** by GPS geofence, timestamped photograph and site ticket. This "proof of disposal" is the design choice that turns a convenience app into a public-health tool: every paid job is a legally disposed job, and the city receives the data.

The project is built on evidence from Kumasi's own ride-hailing market. A 2026 KNUST study of Kumasi drivers found that **89.6% take trips off the app to avoid commissions of 20% or more** and 88.3% feel cheated, and warned that platforms may collapse unless leakage is designed out (Dzisi et al., 2026). KumBola responds with a transparent 15% commission, daily mobile money payouts, prepaid household subscriptions that make off-app dealing pointless, and benefits that exist only on the platform: lease-to-own tricycles, licensing support under the new Road Traffic Regulations, 2026 (L.I. 2519), personal protective equipment (PPE), health screening and National Health Insurance Scheme (NHIS) support. WasteCure's own fleet of 10 tricycles acts as an anchor supply from day one, so households get reliable service before the open marketplace of independent operators reaches scale.

Over 36 months the project will serve **20,000 households (about 72,000 people) and 600 businesses**, support **127 driver livelihoods**, collect **26,195 tonnes** with verified disposal and recover **1,478 tonnes** of recyclables. Operators will earn **GHS 11,584,072**. The platform reaches an operating surplus in **month 27**, earns a **18% operating margin in Year 3**, and exits the grant period with a run-rate surplus of GHS 70,405 a month. We request **GHS 1,725,206 (US$149,368)**, matched by GHS 300,000 in kind and GHS 2,855,115 of reinvested platform revenue. That is a grant cost of **GHS 66 per tonne** collected in the project period, falling to GHS 23 per tonne over five years because no further subsidy is needed.

Before the project is enrolled, the partners will complete **six months of thorough field research**: an operator census, household surveys, cost and price verification, legal review and consultations with the assemblies. During the same six months KoomBei will build a working prototype of the KumBola app, under a separate budget of less than GHS 10,000, and test it with Kumasi households and operators, so the full build starts from validated designs and verified numbers.

WHAT THE GRANT BUYS

Reliable collection for households that already want to pay for it: most Greater Kumasi households are willing to pay more for better tricycle service (Accomford et al., 2025).

Verified disposal: no operator is paid until the load is confirmed at an approved site.

Decent work: fair commission, lease-to-own tricycles, licensing, PPE, health cover and daily pay for informal operators.

City data: a live dashboard for the assemblies showing where waste is generated, collected and dumped.

Sustainable by design: fees, not grants, carry operations from Year 3.

2. The Problem: Greater Kumasi's Waste Emergency

2.1 A fast-growing city generating waste faster than it can manage

The Kumasi Metropolitan Assembly had 443,981 residents at the 2021 census but a daytime population of more than 2.3 million, because the city centre draws commuters from the surrounding municipalities (KMA Composite Budget, 2026). The wider Greater Kumasi Metropolitan Area (GKMA) covers KMA plus 12 districts, about 2.9 million people, growing about 4% a year and projected to reach 4.2 million by 2030 (MobiliseYourCity, 2025). At Ghana's average household size of 3.6 persons (Ghana Statistical Service, 2021), that is roughly 818,056 households.

KMA's own budget uses 1,500 tonnes a day for the metropolis, and the Mayor puts Greater Kumasi at about 2,000 tonnes a day (KMA Composite Budget, 2026; GBC, 2026). Kumasi residents generate about 0.75 kg of waste per person per day, well above the national average of 0.47 kg (Miezah et al., 2015). A 2013 JICA plan projected the Greater Kumasi waste stream would grow from 2,058 tonnes a day in 2013 to 7,685 tonnes a day by 2033 (JICA and LUSPA, 2013).

Indicator	Value	Source
KMA population (2021 census)	443,981	KMA Composite Budget 2026
KMA daytime population	More than 2.3 million	KMA Composite Budget 2026
Greater Kumasi population	About 2.9 million, +4% a year	MobiliseYourCity 2025
Waste generated, Greater Kumasi	About 2,000 t/day	Mayor, via GBC, January 2026
Per capita generation, Kumasi	0.75 kg/person/day	Miezah et al. 2015
Where KMA waste goes	81% Oti landfill, 10% other dumps, 4% burnt, 2% buried	KMA Composite Budget 2026
Households using informal collectors (Suame)	61.5% (formal 35%, dumping 3.5%)	Ampong et al. 2024
Tricycle load factor	0.85 t per trip, 106% of rated capacity	Ampong et al. 2024
Informal collection operating cost	US$36.78 (GHS 432) per tonne	Ampong et al. 2024

Table 1. Greater Kumasi waste at a glance.

2.2 A landfill at its limit

The Oti landfill opened in January 2004 on 40 hectares with a 15-year design life (Owusu-Nimo et al., 2019). It has since deteriorated into what researchers describe as an uncontrolled dumpsite, with 1,125 housing units inside its

500-metre buffer zone (Baah et al., 2024). In 2026 it is receiving 2,000 to 3,000 tonnes a day, above its design capacity. The Mayor warned in January that it would be full within 14 to 16 months without €6 million for two new cells; in April he warned it could close within three months and that KMA would bar other assemblies from using it (MyJoyOnline, 2026a; 2026c; Graphic Online, 2026a). The €20 million Kumasi Compost and Recycling Plant at Adagya, commissioned in 2021 to take pressure off Oti, is idle because of four years of government arrears (Graphic Online, 2026a).

For a household, the consequence is simple: when disposal breaks down, collection breaks down, and waste goes into drains, bushes and fires. In June 2026 tricycle operators spent four days stranded at Oti with full loads and publicly threatened to dump elsewhere (MyJoyOnline, 2026b; Citinewsroom, 2026b). Any collection model that does not track where loads actually go will simply move the problem.

2.3 Health, flooding and economic cost

Waste-choked drains are an official flood driver in Kumasi. In May 2025 NADMO began desilting 17 major drains across Greater Kumasi, blaming filth from households and commercial activity (MyJoyOnline, 2025). In July 2026 the Ashanti Regional Minister said, "We cannot continue to lose lives and properties because our drains are choked with waste" (Citinewsroom, 2026a). Ghana's 2024/25 cholera outbreak passed 6,000 cases and 40 deaths across five regions including Ashanti (Adom Online, 2025). Nationally, poor sanitation has been estimated to cost Ghana US$290 million a year, 1.6% of GDP (World Bank Water and Sanitation Program, 2012).

2.4 The root cause is coordination, not vehicles

Greater Kumasi does not lack tricycles or people willing to collect waste. It lacks the coordination layer that would make them reliable, accountable and bankable. Five failures stand out in the evidence:

- **No reliable booking:** households cannot summon a vetted operator on a schedule; informal fees are negotiated on site and vary (Ampong et al., 2024). Service gaps are worst in low-income areas (Accomford et al., 2025).
- **No proof of disposal:** nobody can verify whether a paid load reached Oti or a drain. KMA's own figures show 16% of metropolitan waste ending in unapproved dumps, burning or burial (KMA Composite Budget, 2026).
- **Unprotected operators:** operators lack PPE and health cover; 43.2% reported coughs and 12.9% carried respiratory pathogens (Armoh et al., 2023). Most have no licence, insurance or pension.
- **Regulation without registration:** KMA branded tricycles in 2018 and banned them from the central business district in 2023, yet riders say the ban has no bye-law behind it (Graphic Online, 2018; GhanaWeb, 2023; GBC, 2025). The city cannot regulate what it cannot see.
- **No data:** there is no recent peer-reviewed figure for Kumasi's collection coverage, and no count of Aboboyaa operators (Adedara et al., 2023). Planning, enforcement and investment happen without evidence.

3. Why the Aboboyaa, and Why Now

3.1 The right vehicle for Kumasi's streets

The Aboboyaa is cheap to buy and run, narrow enough to reach compound houses on unpaved lanes that compactor trucks cannot enter, and simple to repair locally (Fosu, 2025). A new 200cc unit costs GHS 25,000 to 32,000 in 2026 (Yen, 2026). In Kumasi, informal tricycle collectors serve an average of 46 households a day over 43 km (Ampong et al.,

2024). Formal trucks carry more per trip but run at only 62% of capacity and cannot reach many of the households the tricycles serve. The vehicle is not the problem; the absence of a system around it is.

3.2 Why now

- **Legal clarity:** Parliament legalised commercial motorcycles and tricycles in December 2025, and the Road Traffic Regulations, 2026 (L.I. 2519) set clear requirements: riders aged 21 or over, Ghana Card, membership of a commercial transport association, medical and driving tests, DVLA registration, roadworthiness, helmets and reflective wear (Ghana News Agency, 2025; Citinewsroom, 2026c). For the first time there is a formal standard an organised platform can help thousands of operators meet.
- **Landfill pressure:** the Oti crisis gives KMA and the surrounding assemblies a direct interest in tracking tonnage, routing loads to approved sites and cutting illegal dumping.
- **Enforcement is rising:** KMA reintroduced sanitation guards from July 2026, began household sanitation summonses, and prosecuted illegal dumpers in May 2026 (GhanaWeb, 2026; Adom Online, 2026a; Citinewsroom, 2026d). Households now need a legal, affordable way to dispose of waste.
- **Digital and payment readiness:** Ghana has 26.3 million internet users (74.6% penetration) and 41.8 million mobile connections, 96% of them broadband-capable (DataReportal, 2026). The 1% E-Levy on mobile money transfers was repealed in April 2025 (Graphic Online, 2025). 80% of Kumasi taxi drivers already accept mobile money (Dzisi et al., 2026).
- **Demand exists:** most Greater Kumasi households surveyed were willing to pay more for better tricycle service (Accomford et al., 2025). In Accra, residents said they would pay GHS 40 to 50 a month if collection were reliable (Mahama & Ahadzie, 2024).

4. What Ride-Hailing Teaches Us: Evidence from Kumasi

Kumasi is one of the few African cities with peer-reviewed evidence on how ride-hailing platforms actually behave locally. We have used it to design KumBola, copying what works and engineering out what fails.

4.1 What the evidence shows

- 59% of commuters surveyed in Accra and Kumasi had used ride-hailing; users are mostly aged 18 to 39, and 3% needed someone else to book for them (Acheampong et al., 2020). **Implication:** add assisted booking (agents, USSD) for older and non-smartphone users.
- In Kumasi, 82.8% of ride-hail drivers use Bolt; Uber use fell from 64.8% to 13.5% of drivers; drivers multi-home across apps (Dzisi et al., 2026). **Implication:** loyalty must be earned through driver economics, not lock-in.
- Uber, Bolt and Yango retained about 25%, 20% and 15% of fares respectively, regardless of drivers' fuel and data costs (Akorsu, 2023); Kumasi drivers report commissions of 20% or more (Dzisi et al., 2026). **Implication:** a transparent 15% commission.
- 88.3% of Kumasi drivers feel cheated, 89.6% take off-app trips, 84.8% renegotiate fares, and the authors warn platforms may collapse unless leakage is prevented (Dzisi et al., 2026). **Implication:** prepaid subscriptions, daily payouts and platform-only benefits.
- Drivers work 15 or more hours a day to approach the minimum wage, with no enforced rest (Akorsu, 2023); online drivers struck in 2022 over insecurity and 25% charges (Neta et al., 2023). **Implication:** hour caps, fair earnings and a formal voice for operators.

- In Nairobi, 63% of ride-hail users preferred cash and 33% mobile money (Weru & Mugo, 2020). **Implication:** cash payment at agents must remain possible, recorded on the platform.
- India's Recykal, described as "India's Uber of waste collection", tried a household booking app first; poor unit economics forced a pivot to bulk generators (KrAsia, n.d.). **Implication:** do not rely on households alone; build business and institutional revenue from the start.

4.2 Design responses

Failure seen in ride-hailing	KumBola design response
High, opaque commissions (20 to 25%)	15% commission, published to drivers and customers; driver keeps 85% of every fare
Off-app trips and fare haggling	Prepaid household plans; fixed published tariffs; daily mobile money payouts; lease-to-own, insurance and licensing support only for active drivers
Cold start: no drivers, no customers	WasteCure's 10-tricycle anchor fleet guarantees service from launch; WasteCure's existing Kwabre East customers migrate first
Driver exhaustion and insecurity	Daily hour cap, rest prompts, known customers (no blind requests), SOS button, association engagement
Cash leakage	Cash accepted only at registered agents and logged on the platform
Single-segment dependence (Recykal lesson)	Three revenue streams from Year 1: households, businesses, recyclables; institutional contracts from Year 2
Regulatory ambiguity	Compliance by design: L.I. 2519 checks at onboarding, KMA and assembly MoUs, data sharing with the city

Table 2. Lessons from ride-hailing translated into KumBola design decisions.

5. The Solution: How KumBola Works

KumBola is a three-sided platform connecting waste generators (households, businesses and institutions), Aboboyaa operators, and the city. KoomBei builds and runs the technology; WasteCure runs operations on the ground.

5.1 The service in five steps

1. Book	2. Dispatch	3. Collect	4. Verify disposal	5. Pay
Customer books by web app, USSD (*code#), WhatsApp or a community agent; chooses a plan or one-off pickup	Platform assigns the nearest vetted operator, weighted by rating and route; customer sees name, photo, plate and arrival time	Operator collects, confirms with a pickup code or photo; segregated recyclables are logged by weight	Operator unloads at an approved site; GPS geofence, timestamped photo and site ticket close the job	Driver is paid 85% by mobile money the same day; customer gets a receipt and disposal confirmation; city dashboard updates

Table 3. The KumBola job cycle. A job is not complete, and the operator is not paid, until disposal is verified.

5.2 Customer channels and plans

Households choose from four plans (Section 8). Booking works on a smartphone web app (installable, low-data), by USSD on any phone, by WhatsApp, or through community agents at WasteCure collection points and local shops. Payment is prepaid by MTN MoMo, Telecel Cash or AirtelTigo Money through a licensed aggregator (Paystack or Hubtel), with cash accepted at registered agents. Businesses and institutions receive monthly invoices and a dashboard of their collections and tonnages.

5.3 The operator side

Operators join through a structured onboarding: Ghana Card and licence checks aligned with L.I. 2519, vehicle roadworthiness and covered cargo bed, a one-day safety, customer service and app training, a PPE kit, and enrolment in NHIS. They receive a driver app (offline-capable), daily mobile money payouts, a recyclables bonus for clean segregated loads, and access to a lease-to-own tricycle fund. Performance is managed through ratings, on-time rates and disposal compliance, with a transparent appeals process.

5.4 Proof of disposal: the core innovation

Every KumBola job ends at an approved disposal or transfer site. The driver app records arrival inside a GPS geofence around the site, a timestamped photograph of the unload, and the site's ticket or weighbridge record where available. Payment is released only when these match. Random audits, device checks against GPS spoofing, and cross-checks between collected and disposed tonnages deter fraud. If Oti is closed or disrupted, the dispatch engine routes loads to the next approved site designated by the relevant assembly or to WasteCure's aggregation point. This single rule converts private convenience into a public good: the city gets a verified record of where waste goes, and illegal dumping stops being profitable for platform operators.

5.5 Recyclables and the circular economy

Ghana's municipal waste is about 61% organic and 14% plastic by weight (Miezah et al., 2015). Households on KumBola receive colour-coded sacks and simple segregation guidance. Operators deliver clean plastics, metals and paper to WasteCure's upgraded aggregation point, where they are weighed, baled and sold to processors. Operators earn a per-kilogram bonus. The recovered plastic tonnage also creates an option to register for plastic credits through an established developer; Environment360 and GreenCollar run a registered Verra plastic project in Ghana with planned expansion to Kumasi (GreenCollar, n.d.). Plastic credits are treated as upside only and are excluded from the base-case financials.

5.6 City dashboard

KMA and partner assemblies receive a free dashboard (with a paid service tier for contracted zones) showing live collection heatmaps, tonnages by area, disposal site usage, operator registration and compliance, and geotagged reports of illegal dumping submitted by customers and drivers. This gives the assemblies the evidence base that is currently missing for planning, enforcement and landfill management.

5.7 Technology architecture

Layer	Choice	Why
Customer and driver apps	Next.js progressive web apps, offline-first; native Android and iOS in Year 2	Low data use on mid-range Android phones; one codebase
Backend and database	Supabase (PostgreSQL, authentication, storage) on a managed cloud region	Managed and secure; row-level security for personal data; hosting region reviewed against Act 843
Payments	Paystack or Hubtel mobile money collections and payouts (1.95% fee)	Licensed aggregators, all three networks, daily payouts
USSD and SMS	Shared USSD code via a licensed aggregator; bulk SMS	Feature-phone access and booking confirmations
Dispatch	Distance, rating and route-aware matching; geofenced disposal verification	Reliability and proof of disposal

Layer	Choice	Why
Monitoring	Error monitoring, uptime alerts, audit logs	Operational resilience and fraud control

Table 4. Technology stack. Built by KoomBei Digital Limited, web first, starting from the prototype tested during the pre-enrolment research.

6. Partnership, Governance and Team

6.1 Roles

Partner	Role and contribution
KoomBei Digital Limited (Accra)	Technology lead: product design, platform build and maintenance, data and dashboard, payments integration, data protection compliance. Contributes 30% of the platform build in kind
WasteCure Limited (Kumasi, founded 2021)	Implementation lead: operations, driver recruitment and supervision, customer acquisition, aggregation point and recyclables sales, assembly relationships. Contributes its anchor fleet of 10 tricycles, existing customers and site
KMA and Suame, Kwabre East, Oforikrom assemblies (proposed MoUs)	Recognition of KumBola operators, access to approved disposal sites, data sharing, co-design of the dashboard, potential service contracts
Sanitation Tricycle Riders Association (proposed)	Operator engagement, grievance handling, representation on the operator council
KNUST research team (proposed)	Independent baseline, mid-term and endline evaluation; the Kumasi evidence base cited here is largely KNUST's

6.2 Governance

The partners will form a jointly owned special purpose company, KumBola Limited, governed by a written joint venture agreement signed before any grant funds are disbursed. The agreement will set out equity, decision rights, intellectual property (platform code licensed to the joint venture, with KoomBei retaining ownership of its reusable components), data ownership (customer and operator data held by the joint venture as data controller under the Data Protection Act, 2012 (Act 843)), revenue sharing, and exit terms. A Project Steering Committee of both partners, an assembly representative and an operator representative will meet quarterly. Grant funds will be held in a dedicated account with dual signatories and audited annually.

6.3 Core team

Name	Role	Relevant experience
Frank Koomson	Project Director and technology lead (KoomBei)	Founder of KoomBei Digital, building web platforms for NGOs and start-ups; backend web development training (MEST Africa, 2024); BA Geography and Political Science (University of Ghana); founder of the youth nonprofit SWK Ghana
George Adomako	Operations lead (WasteCure)	Founder and Executive Director of WasteCure; built plastic collection and door-to-door services from Kwabre East since 2021
Maxwell Opoku Amoako	Field operations and fleet (WasteCure)	Technical and Operations Manager, WasteCure
Edmond Amankwaah	Customer acquisition and partnerships (WasteCure)	Sales, Marketing and Product Development, WasteCure
To be recruited	Operations manager, developer, field officers, M&E officer	Staffing plan in the financial model: 6 staff in Year 1 rising to 13 in Year 3

Table 5. Core team. WasteCure roles as listed on wastecure.org; to be confirmed by WasteCure before submission.

7. Target Area, Beneficiaries and Market

7.1 Pilot municipalities

Municipality	Population (2021)	Why it was chosen
Suame	136,290	Best-documented tricycle market in Kumasi: 61.5% of households use informal collectors (Ampong et al., 2024), giving a strong baseline
Kwabre East	296,814	WasteCure's home base since 2021; existing customers and collection site for the anchor fleet
Oforikrom	213,126	Dense student and middle-income housing around KNUST; high smartphone and mobile money use
Year 2 expansion: Asokwa, Old Tafo, Kwadaso	394,536	Adjacent areas with heavy tricycle use; Asokwa collectors already organised (Adom Online, 2026b)

Table 6. Target municipalities. Populations are 2021 census figures as reported by secondary sources, to be confirmed against Ghana Statistical Service tables during inception.

7.2 Beneficiaries by month 36

- **20,000 households, about 72,000 people**, with reliable, affordable, verified collection, including lower-cost compound and pay-as-you-go plans.
- **600 small businesses** (chop bars, shops, hostels, schools, churches) and **3 institutional clients** (assembly zones, markets or estates).
- **127 Aboboyaa operators** earning fair, regular income with licences, PPE, health cover and a path to vehicle ownership; **13 staff** and about 40 part-time community agents.
- **The assemblies and residents city-wide**, through less open dumping and burning, fewer choked drains, and better data for landfill and sanitation planning.

7.3 Market size

Level	Definition	Households	Annual value (GHS)
Total addressable market	All Greater Kumasi households at the blended KumBola tariff	818,056	408,373,333
Serviceable market	Households in the six target municipalities that rely on informal or tricycle collection (61.5%)	177,798	88,756,524
Target by month 36	Active household accounts (11.2% of the serviceable market)	20,000	9,984,000

Table 7. Market sizing using a blended household tariff of GHS 41.60 a month. At month 36 KumBola handles about 67 tonnes a day, roughly 3.4% of Greater Kumasi's waste: a credible, deliberately modest share.

8. Business Model and Pricing

8.1 Revenue streams

Stream	How it works	3-year revenue (GHS)	Share
Commission	15% of every household, business and institutional booking	2,044,248	72%

Stream	How it works	3-year revenue (GHS)	Share
Recyclables margin	Sale of segregated recyclables through WasteCure, net of driver bonus and handling	738,867	26%
Dashboard and data services	Fee per institutional contract for reporting and proof-of-disposal data	72,000	3%
Plastic credits	Upside only, not in base case	0	0%
Total		**2,855,115**	**100%**

8.2 Household tariffs

Plan	Price (GHS/month)	Pickups/ month	Share of accounts	Who it serves
Individual weekly	45	4.3	50%	Most households: one 240-litre bin or three sacks a week
Compound shared (per household)	30	2.2	25%	Compound houses sharing a bin, twice-weekly collection
Premium twice-weekly	80	8.7	10%	Larger or higher-income households
Pay-as-you-go small sack	About 24 (GHS 6 per sack)	4.0	15%	Low-income and irregular users; no commitment

The blended tariff is GHS 41.60 per household a month, or about GHS 9.97 per pickup. This is about 65% below app-based pricing in Accra, where Jumeni charges GHS 30 a week for a 240-litre bin and GHS 40 to 50 for a one-off pickup (Jumeni, 2026), and within the range Accra households said they would pay for reliable service (Mahama & Ahadzie, 2024). The compound and pay-as-you-go plans keep the service within reach of low-income households: a compound plan costs a household GHS 1 a day. Businesses pay an average of GHS 300 a month for three pickups a week. Institutional contracts are priced at about GHS 15,000 a month per zone plus a GHS 2,000 dashboard fee.

8.3 Why the model holds together

Each pickup generates about GHS 9.97. The operator keeps 85%, which at 30 platform pickups a day yields net earnings of GHS 3,296 a month for an owner-operator after fuel, tolls, maintenance and vehicle costs. The platform keeps 15% plus the recyclables margin, and its variable costs (payments, SMS, payouts, driver welfare) absorb only about 17% of revenue. The remaining contribution covers a lean fixed team. The key operating risks are the cost of acquiring households and the price of recyclables, which is why both are tested in the sensitivity analysis (Section 9.5) and verified in the field during the 6-month pre-enrolment research (Annex B).

9. Financial Projections and Profit Margins

The projections come from a 36-month, formula-driven model submitted with this proposal (KumBola_Financial_Model.xlsx). All figures are in constant September 2026 cedis; prices and costs are assumed to move together. Every assumption, its source and a confidence rating is listed in the model and summarised in Annex A.

9.1 Three-year summary (base case)

GHS	Year 1	Year 2	Year 3	Total
Household accounts (end of year)	2,500	9,000	20,000	20,000
Active drivers (end of year)	15	56	127	127
Tonnes collected, verified disposal	814	6,952	18,428	26,195
Gross booking value (paid by customers)	445,920	3,682,400	9,500,000	13,628,320
Paid to drivers and fleet owners	379,032	3,130,040	8,075,000	11,584,072
Commission revenue	66,888	552,360	1,425,000	2,044,248
Recyclables net margin	12,215	173,809	552,843	738,867
Dashboard and data fees	0	12,000	60,000	72,000
Total platform revenue	**79,103**	**738,169**	**2,037,843**	**2,855,115**
Staff	222,500	444,000	552,000	1,218,500
Customer acquisition	73,925	234,325	446,300	754,550
Driver onboarding, kits and welfare	28,320	92,160	197,000	317,480
Payments, payouts, SMS, USSD, hosting	30,737	141,254	329,211	501,203
Office, transport, compliance, education, contingency	81,824	114,887	145,526	342,237
Total operating costs	**437,306**	**1,026,626**	**1,670,037**	**3,133,969**
Operating surplus / (deficit)	**(358,203)**	**(288,458)**	**367,806**	**(278,854)**
Operating margin	n/m	-39%	18%	-10%

Table 8. Base-case three-year summary. n/m = not meaningful (Year 1 is a build and pilot year). Driver onboarding row includes kits, training and welfare; the payments row includes payment processing, payouts, SMS, USSD and hosting; the office row includes contingency.

9.2 Path to break-even

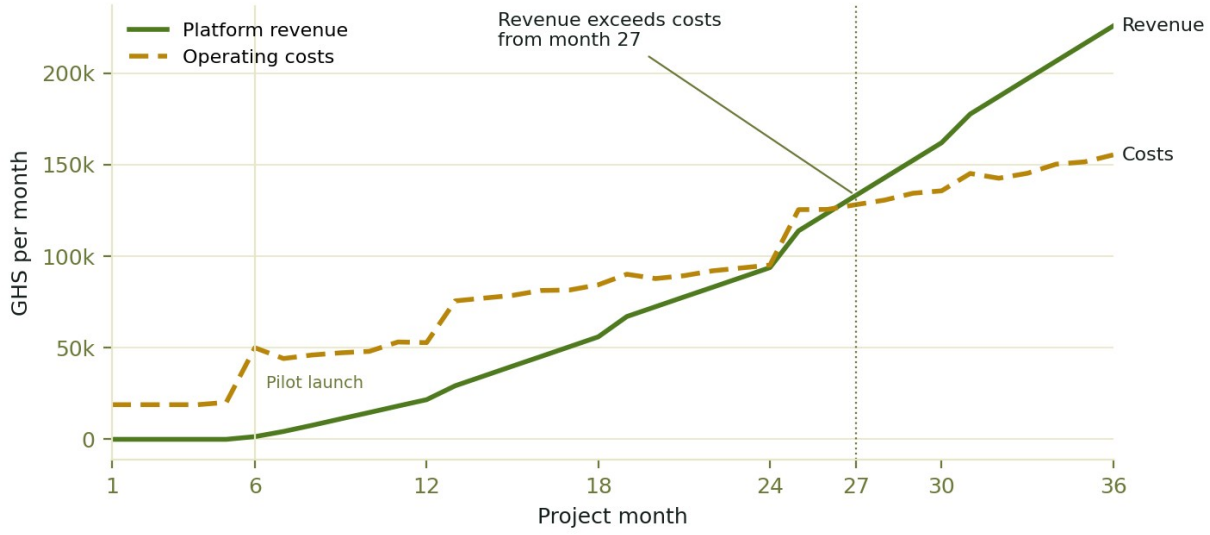


Figure 1. Monthly platform revenue and operating costs, base case. Revenue overtakes costs in month 27.

Year 1 is an investment year: five months of inception and build, then a seven-month pilot. Costs step up at launch and again at the start of Years 2 and 3 as the team grows. Revenue exceeds operating costs from **month 27**, and by month 36 the platform earns GHS 225,750 a month against costs of GHS 155,345, a monthly surplus of **GHS 70,405 (31% margin)**.

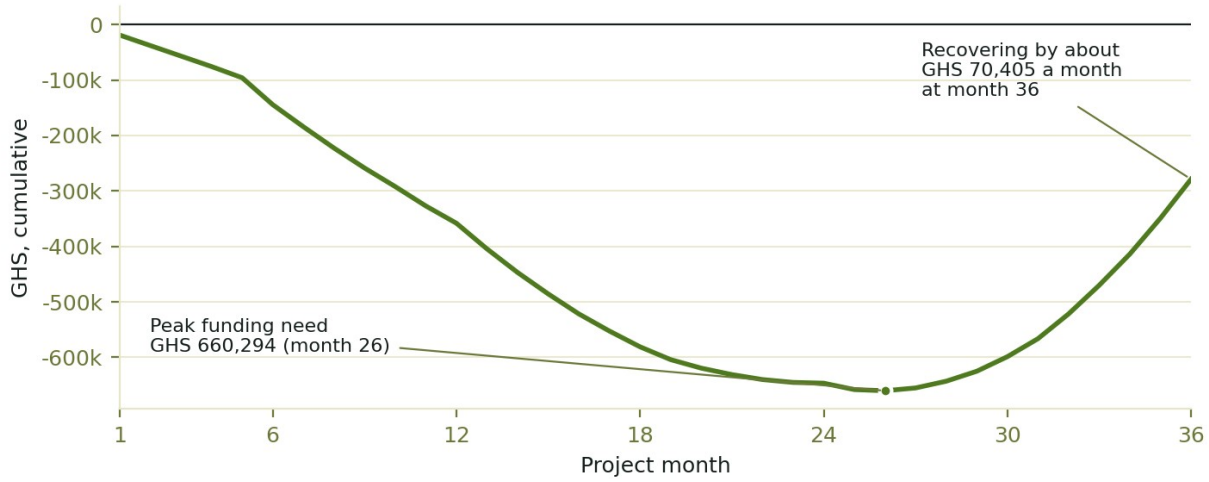


Figure 2. Cumulative operating position. The deepest point, GHS 660,294, is the operating subsidy the grant must cover; it is included in the budget.

9.3 Profit margins

Margin measure	Year 1	Year 2	Year 3	Month 36 run-rate
Contribution margin (revenue less variable costs)	77%	82%	83%	About 83%
Operating margin (surplus / platform revenue)	n/m	-39%	18%	31%
Platform revenue as share of customer payments	18%	20%	21%	
Operating surplus as share of customer payments	n/m	-8%	4%	7%

Most of the value created flows to operators, not to the platform: of every GHS 100 a customer pays, GHS 85 goes to the driver. In Year 3 the platform keeps about GHS 4 as operating surplus. This is deliberate. A thin, transparent platform margin is what keeps operators on the app (Section 4), and at scale it is still enough to fund maintenance, upgrades and expansion without further grants. Surpluses will be reinvested in expansion to new municipalities and the lease-to-own fund; the joint venture will not distribute dividends during the grant period.

9.4 Driver economics

GHS per working day	Owner-operator	Lease-to-own (first 24 months)
Platform pickups per day	30	30
Fare earnings (85% of fares)	254.11	254.11
Recyclables bonus	6.74	6.74
Fuel (43 km at 18 km/L, GHS 16.77/L)	(40.06)	(40.06)
KMA daily fee and landfill tolls	(40.00)	(40.00)
Maintenance, tyres, phone, insurance, licence	(34.81)	(34.81)

GHS per working day	Owner-operator	Lease-to-own (first 24 months)
Vehicle cost (depreciation or lease repayment)	(19.23)	(57.69)
Net daily earnings	**126.75**	**88.29**
Net monthly earnings (26 days)	**3,296**	**2,296**
Multiple of national minimum wage (GHS 21.77/day)	5.8x	4.1x

Table 9. One operator's daily economics, base case, Year 2 recyclables share. Kumasi ride-hail drivers most commonly earn GHS 1,001 to 1,500 a month (Dzisi et al., 2026).

9.5 Scenarios and sensitivity

Case (one change from base)	Households M36	Y3 revenue	Y3 surplus	Y3 margin	First surplus month	Peak funding need	Grant needed
Base case	20,000	2,037,843	367,806	18%	27	660,294	1,725,206
Conservative: adoption x0.6, prices -10%, 25 pickups/driver	12,000	1,228,126	(74,889)	-6%	32	898,266	1,979,836
Upside: adoption x1.3, 34 pickups/driver, plastic credits	26,000	2,945,206	1,022,394	35%	20	521,166	1,576,339
Commission 12%	20,000	1,752,843	82,806	5%	30	842,016	1,919,648
Commission 18%	20,000	2,322,843	652,806	28%	22	546,646	1,603,603
Household prices 20% lower	20,000	1,813,827	174,369	10%	29	779,860	1,853,141
No recyclables revenue	20,000	1,485,000	(185,037)	-12%	34	1,033,158	2,124,170
Acquisition cost doubled (GHS 50)	20,000	2,037,843	(53,664)	-3%	32	1,093,177	2,188,391
No institutional contracts	20,000	1,856,343	215,617	12%	28	709,823	1,778,202

Table 10. Each row is a full re-run of the financial model (GHS). Base case in the first row.

The model is robust to most single shocks: even at 60% of target adoption and 10% lower prices, the platform reaches a monthly surplus by month 32. Two variables matter most. **Recyclables revenue**: without it, surplus is delayed to month 34. **Customer acquisition cost**: doubling it delays surplus to month 32. Both are addressed directly. Recyclables prices will be fixed through offtake agreements with Kumasi aggregators and processors before launch, and the recyclables share assumed is conservative (3% of tonnage in Year 1 rising to 6%, against a plastics share of about 14%). Acquisition cost is kept low by migrating WasteCure's existing customers, compound-house plans that sign up several households at once, referral credits, and assembly endorsement. Expansion spending is gated on these metrics (Section 12), so a weaker case slows growth rather than exhausting funds.

10. Budget and Grant Request

Budget line (GHS)	Year 1	Year 2	Year 3	Total	Grant	In kind
A. Technology platform (KoomBei)						
Platform MVP: customer web app, driver PWA, dispatch engine, MoMo, USSD, proof-of-disposal, admin	150,000	0	0	150,000	105,000	45,000
KMA/assembly data dashboard and open data reporting module	30,000	0	0	30,000	20,000	10,000
Native Android/iOS apps, offline mode, e-trike telemetry	0	90,000	0	90,000	70,000	20,000

Budget line (GHS)	Year 1	Year 2	Year 3	Total	Grant	In kind
B. Fleet, safety and livelihoods (WasteCure)						
Anchor fleet: 10 WasteCure tricycles committed to the platform (in kind, at used value)	200,000	0	0	200,000	0	200,000
Lease-to-own revolving fund: 10 new 200cc tricycles for independent riders	300,000	0	0	300,000	300,000	0
E-trike pilot: 4 electric cargo tricycles + solar charging point	0	182,048	0	182,048	182,048	0
Rider licensing support under L.I. 2519 (medical, tests, licence) for 50 riders	10,000	10,000	0	20,000	20,000	0
C. Recycling and segregation						
Aggregation point upgrade at WasteCure: shed, platform scales, manual baler	90,000	0	0	90,000	70,000	20,000
Colour-coded starter sacks for first 3,000 households	24,000	0	0	24,000	24,000	0
D. Research, M&E and engagement						
Baseline and endline studies with a KNUST research partner (400 households, 60 operators, GIS)	45,000	0	35,000	80,000	80,000	0
Stakeholder engagement: KMA, assemblies, Tricycle Riders Association, traditional authorities	15,000	10,000	5,000	30,000	30,000	0
Independent annual audit	12,000	12,000	12,000	36,000	36,000	0
Company set-up, JV agreement, data protection registration	20,000	0	0	20,000	15,000	5,000
E. Operations						
Operating deficit cover to the peak cumulative deficit (from model; Year 3 funded by platform revenue)	358,203	302,091	0	660,294	660,294	0
F. Indirect costs						
Project management and indirect costs (7% of grant-funded direct costs)	68,194	41,030	3,640	112,864	112,864	0
Total project cost	**1,322,397**	**647,169**	**55,640**	**2,025,206**	**1,725,206**	**300,000**
Total in US$ (GHS 11.55 = US$1)	**114,493**	**56,032**	**4,817**	**175,343**	**149,368**	**25,974**

Table 11. 36-month budget. In kind: KoomBei software development (GHS 80,000) and WasteCure's anchor fleet and aggregation site (GHS 220,000).

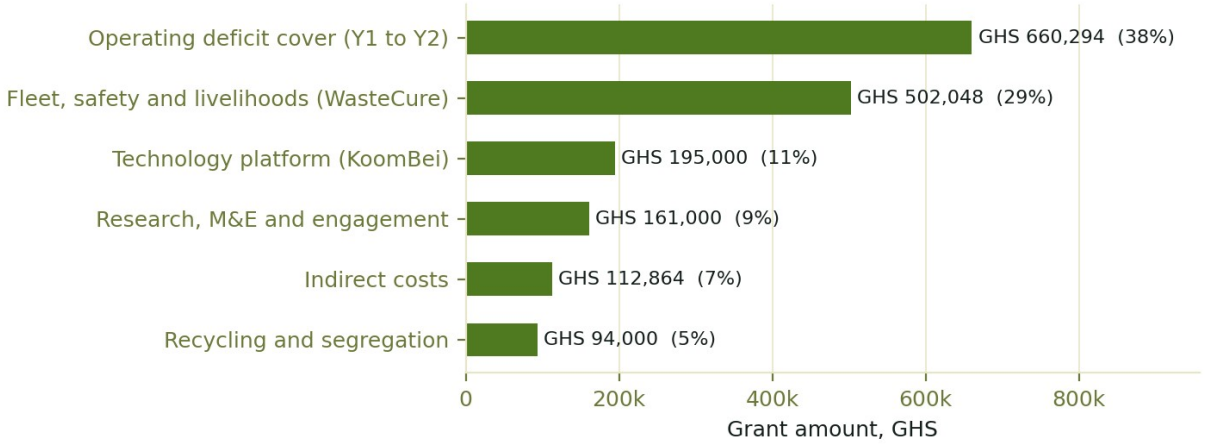


Figure 3. How the grant is used, by category.

10.1 Notes on the budget

- **Grant tapers to zero:** the grant is front-loaded: GHS 1,042,397 in Year 1, GHS 627,169 in Year 2 and only GHS 55,640 in Year 3 (audit, endline evaluation, engagement). Operations in Year 3 are funded by platform revenue.
- **Revolving fund:** the GHS 300,000 lease-to-own fund buys 10 new tricycles for independent operators. Each operator repays GHS 36,000 over 24 months; GHS 360,000 is recovered by month 36 and recycled into further tricycles, so the fund keeps growing after the grant ends.
- **E-trike pilot:** four electric cargo tricycles (Solar Taxi, GHS 39,262 each) with a solar charging point test whether electric operation lowers fuel costs and emissions on Kumasi routes, informing a larger transition.
- **Operating deficit cover:** sized to the model's peak cumulative deficit of GHS 660,294, not to annual totals, so the project cannot run out of cash mid-year.
- **Research phase and prototype:** the 6-month pre-enrolment research and the KumBola app prototype (under GHS 10,000, set out in a separate KoomBei proposal) are resourced separately and are not part of this 36-month budget.
- **Modular funding:** the budget can be funded in modules: platform and pilot operations; fleet and livelihoods (revolving fund, e-trikes, licensing); and recycling, research and engagement. This allows co-funding by several funders (Annex C).

11. Value for Money

Dimension	Evidence
Economy	Lean team (6 staff in Year 1, 13 in Year 3); Kumasi salary levels; no vehicles bought for the platform itself; open-source and managed cloud tools at about US$45 a month at launch; 7% indirect cost rate; 15% in-kind co-financing
Efficiency	Grant cost of GHS 66 per tonne collected with verified disposal over 36 months and GHS 23 per tonne over five years, compared with an informal collection operating cost of GHS 432 per tonne (Ampong et al., 2024). Grant cost of GHS 86 per household served
Effectiveness	26,195 tonnes with verified disposal; 1,478 tonnes of recyclables; about 2,619 tCO2e avoided; 20,000 households and 600 businesses served; platform self-financing from month 27
Equity	Compound and pay-as-you-go plans for low-income households; USSD and agent booking for feature-phone users; GHS 11,584,072 of income to informal operators (6.7 cedis per cedi of grant); grant cost of GHS 12,323 (US$1,067) per livelihood
Leverage	Every cedi of grant is matched by 1.83 cedis of in-kind contribution and reinvested platform revenue during the project

Table 12. Value for money against the four Es, plus leverage.

12. Implementation Plan and Milestones

Phase	Months	Key activities	Deliverables and stage gate
Pre-enrolment research	6 months before month 1	Thorough field research: operator census and interviews, KNUST household baseline survey (400 households, 60 operators, GIS map), price and cost verification, legal review of L.I. 2519, consultations with KMA, assemblies and the Riders Association; KoomBei builds and field-tests the app prototype (separate budget, under GHS 10,000)	Research report; validated assumptions (Annex B); tested prototype; partners' go or no-go decision to enrol

Phase	Months	Key activities	Deliverables and stage gate
0. Inception	1 to 2	Joint venture agreement; assembly MoUs; confirm research findings and update the financial model; Data Protection Commission registration	Signed JV and MoUs; updated model and budget
1. Design and build	2 to 5	User research and prototype; platform build (web apps, dispatch, mobile money, USSD, proof of disposal, dashboard); recruit and license first 20 operators; upgrade aggregation point	Gate 1: MVP passes field testing on mid-range Android and feature phones; 20 operators licensed under L.I. 2519
2. Pilot	6 to 12	Launch in Kwabre East and Suame (month 6), Oforikrom (month 9); migrate WasteCure customers; business accounts from month 8; 10 lease-to-own tricycles	Gate 2 (month 12): at least 2,000 active households, 90% on-time pickups, 95% verified disposal, 80% customer satisfaction
3. Scale	13 to 24	Expand to Asokwa, Old Tafo, Kwadaso; native apps; e-trike pilot (month 15); first institutional contract (month 19); mid-term evaluation (month 18)	Gate 3 (month 24): 9,000 households; contribution margin above 75%; first contract live
4. Consolidate	25 to 36	Contracts 2 and 3; plastic credit registration; endline evaluation; replication plan for Sunyani and other secondary cities	Operating surplus from month 27; 20,000 households; endline report and replication plan

Table 13. Implementation phases. The project is enrolled only after the 6-month research phase; expansion spending is released only when the previous stage gate is met.

13. Theory of Change and Results Framework

If Aboboyaa operators are organised on a platform that guarantees them fair, regular pay, safety and a legal status, **and** households can book and pay for reliable collection in the way they already book transport, **and** every paid job must end at an approved site, **then** more waste will be collected and legally disposed of, operators will earn decent incomes, and the city will gain the data to manage its waste system, **so that** Greater Kumasi becomes cleaner, healthier and less flood-prone, through a service that pays for itself.

Level	Result	Indicator	Baseline	Target (M36)	Source of verification
Impact	Cleaner, healthier Greater Kumasi	Share of households in served areas disposing by burning or dumping	To be set at baseline	Down 50% in served areas	KNUST baseline and endline surveys
Outcome 1	Reliable, verified collection	Active household accounts; on-time pickup rate; share of loads with verified disposal	0; n/a; n/a	20,000; 90%; 95%	Platform data; audits
Outcome 2	Decent livelihoods for operators	Active operators; average net monthly earnings; NHIS enrolment; L.I. 2519 licensing	0; baseline survey; baseline	127; above GHS 2,300; 100%; 100%	Platform payout records; operator survey
Outcome 3	More resources recovered	Tonnes of recyclables sold; share of households segregating	0; baseline	1,478 t; 40%	Aggregation point weigh records; survey
Outcome 4	Better city data and governance	Assemblies using dashboard; illegal dumping reports resolved	0	4 assemblies; monthly reporting	MoUs; dashboard logs
Outcome 5	Financial sustainability	Month of first operating surplus; Year 3 operating margin	n/a	Month 27; 18%	Audited accounts
Outputs	Platform, fleet and capacity	Platform live; operators trained with PPE; lease-to-own tricycles; e-trikes; aggregation point	0	Live by month 5; 127 trained; 10 plus; 4; 1 site	Project records

Table 14. Results framework. Baselines marked for the survey will be measured during the 6-month pre-enrolment research.

14. Monitoring, Evaluation and Learning

Because every job passes through the platform, most indicators are measured automatically and in real time: bookings, pickups, on-time rates, tonnages, disposal verification, operator earnings, payouts and complaints. A monthly dashboard will be shared with the Steering Committee and funders, and quarterly reports will combine platform data with field checks.

- **Independent evaluation:** a KNUST team will run a baseline (during the 6-month pre-enrolment research), mid-term review (month 18) and endline (months 34 to 36), comparing served and similar unserved neighbourhoods on disposal practice, satisfaction, operator income and health.
- **Data quality:** random audits of 5% of disposal verifications each month; reconciliation of collected and disposed tonnages; GPS spoofing detection.
- **Learning questions:** Which plan mix gives low-income households reliable service at acceptable cost? Does proof of disposal change dumping behaviour in served areas? Do platform benefits reduce off-app dealing compared with ride-hailing? Is electric operation viable on Kumasi routes?
- **Sharing:** findings published openly for KMA, the Ministry of Sanitation and other cities; anonymised, aggregated data shared with the assemblies under the MoUs.

15. Benefits and Impact

Who benefits	Benefit	Scale (36 months)
Households	Reliable, scheduled collection; pay only for completed pickups; lower-cost shared and pay-as-you-go plans; proof that waste was disposed of legally	20,000 households, about 72,000 people
Aboboyaa operators	Regular, transparent income; daily payouts; licensing, PPE, health screening and NHIS; path to owning a tricycle; a voice through the association	127 operators; GHS 11,584,072 earned
Businesses and institutions	Scheduled collection with records for compliance; data on waste volumes	600 businesses; 3 contracts
Assemblies and KMA	Verified disposal records; heatmaps; illegal dumping reports; a registered, visible operator base; pressure off landfill through recycling	4 or more assemblies
Environment and climate	Less open burning and dumping; recyclables recovered; e-trike trial	1,478 t recycled; about 2,619 tCO2e avoided (estimate)
Public health	Fewer choked drains and breeding sites; safer working conditions for operators	Measured through the baseline and endline surveys
Local economy	13 staff jobs, about 40 part-time agents, business for mechanics, tricycle dealers and recyclers	140 direct livelihoods

15.1 Alignment with policy and the Sustainable Development Goals

Framework	How KumBola contributes
SDG 11.6 (city waste), 3.9 (pollution and health), 6.3 (water quality)	More waste collected and legally disposed; fewer drains blocked
SDG 8.3 and 8.8 (decent work, safe working environments)	Formalised, protected livelihoods for informal operators

Framework	How KumBola contributes
SDG 12.5 (reduce waste through recycling) and 13 (climate)	Recyclables recovery; avoided open burning; e-trike trial
Local Governance Act, 2016 (Act 936)	Supports assemblies' legal duty for waste management with data and registered operators
Road Traffic Regulations, 2026 (L.I. 2519)	Onboarding checks help operators meet licensing, association and safety requirements
Environmental Protection Act, 2025 (Act 1124) and the draft Extended Producer Responsibility law	Verified tonnage and recyclables data suitable for future producer responsibility schemes and the Ghana Carbon Registry
National Plastics Management Policy (2019) and Ghana NPAP	Source segregation and plastic recovery in a city outside Accra
Data Protection Act, 2012 (Act 843)	Registered data controller; privacy by design

16. Sustainability and Exit Strategy

- **Financial:** customer fees, not grants, fund operations from Year 3. The platform reaches an operating surplus in month 27 and exits the grant with a run-rate surplus of GHS 70,405 a month, about GHS 844,862 a year at month-36 scale before any further growth. The revolving fund recovers GHS 360,000 by month 36 and keeps financing tricycles.
- **Institutional:** a jointly owned company with a clear agreement, audited accounts and a steering committee; MoUs with the assemblies that make KumBola part of the city's system rather than a parallel one; operator representation through the association.
- **Technical:** a platform owned by the joint venture, built on maintained, affordable cloud services, with documentation and a second developer by Year 2 so the system does not depend on one person.
- **Environmental:** segregation habits, recyclables offtake agreements and, if viable, plastic credits create a lasting circular-economy stream; the e-trike pilot builds the case for cleaner vehicles.
- **Social:** operators gain licences, health cover, skills and, through lease-to-own, assets they keep. Households gain a service they value enough to pay for.
- **Replication:** the platform is city-agnostic. The endline will produce a replication plan for Sunyani and other secondary cities, financed by surpluses and follow-on investment rather than new grants.

17. Risk Register

Risk	L	I	Mitigation	Owner
Household adoption slower than planned	M	H	Migrate WasteCure customers first; compound plans; referral credits; assembly endorsement; stage-gated expansion	WasteCure
Customer acquisition costs higher than assumed	M	H	Community agents paid on results; bulk sign-up through compounds, churches and schools; track cost per account monthly	WasteCure
Recyclables prices lower or volatile	M	M	Offtake agreements before launch; conservative share assumed; plastic credits as upside only	WasteCure
Operators deal off the platform	H	M	Low commission; prepaid plans; daily pay; platform-only benefits; published fares	Both

Risk	L	I	Mitigation	Owner
Oti closure or disruption; KMA bars other assemblies	H	H	Dispatch reroutes to approved alternative sites; assembly MoUs naming approved sites; WasteCure transfer point	Both
Disposal fraud (fake photos, GPS spoofing)	M	H	Geofence plus photo plus site ticket; device checks; 5% random audits; payment clawback and suspension	KoomBei
Regulatory friction (CBD ban, tolls, L.I. 2519 cargo rules)	M	M	Early legal review; MoUs; operators routed outside restricted zones; pay tolls through the platform	Both
Competitor response (Zoomlion, Accra apps entering Kumasi)	M	M	Open-marketplace model and assembly partnerships as moat; speed in pilot zones	Both
Fuel price shocks	M	M	Tariff indexation clause; route optimisation; e-trike pilot	Both
Operator health and road safety incidents	M	H	PPE, training, helmets, hour caps, insurance, health screening, SOS feature	WasteCure
Data breach or misuse	L	H	Act 843 registration; encryption; role-based access; minimal data collection; incident plan	KoomBei
Platform outage or poor connectivity	M	M	Offline-capable apps; USSD fallback; monitoring and alerts; manual dispatch backup	KoomBei
Displacement of informal collectors	L	H	Onboard existing operators rather than replace them; no exclusivity claims over any area	Both
Partner disagreement	L	H	Signed JV agreement with IP, data and exit terms; steering committee; mediation clause	Both

Table 15. Risk register. L = likelihood, I = impact (H high, M medium, L low).

18. Safeguarding, Inclusion, Health and Data Protection

- **Operator health and safety:** every operator receives PPE (gloves, masks, boots, reflective vest, helmet), a cargo cover to stop spillage, safety training and annual health screening, responding directly to the pathogen findings of Armoh et al. (2023). Daily hour caps and rest prompts apply.
- **Child labour:** operators must be 21 or older in line with L.I. 2519; no one under 18 may work on a KumBola job, including as a helper.
- **Gender inclusion:** women are recruited as community agents, sorters, customer support staff and, with training, operators; the e-trike, being lighter and quieter, is trialled partly for this purpose. Sex-disaggregated data will be reported.
- **Low-income and digital inclusion:** USSD, WhatsApp and agent booking; cash at registered agents; compound and pay-as-you-go plans; Twi-language support.
- **Grievance mechanism:** a free-to-use complaints channel for customers, operators and residents, with response within 24 hours and escalation to the steering committee.
- **Data protection:** the joint venture will register with the Data Protection Commission, collect only what is needed, encrypt personal data, publish a plain-language privacy notice, and share only aggregated or anonymised data with the assemblies.

19. Competitive Landscape

Operator	Model	How KumBola differs
Zoomlion Ghana and KMA contractors	Franchise and truck-based collection; Zoomlion app and USSD for bin requests and payments since 2021 (MyJoyOnline, 2021)	Tricycle-first, reaches lanes trucks cannot; open to independent operators; proof of disposal
Informal Aboboyaa operators	Cash, negotiated fees, no booking, tracking or verification	KumBola organises them rather than competing with them
Jumeni (Accra)	App booking with its own verified collectors; GHS 30 a week; wallet balance required from 2026 (Jumeni, 2026)	Not in Kumasi; closed collector pool; KumBola is an open marketplace with lower Kumasi pricing
Borla Commando (Accra)	On-demand pickups with its own branded tricycle fleet (Borla Commando, 2026)	Closed fleet; KumBola uses the existing Kumasi fleet and adds verified disposal
AfriNova Wastle (national, launched July 2026)	Digital waste and flood-prevention platform (Citinewsroom, 2026e)	KumBola is operator-centred with payments, dispatch and disposal verification in Kumasi

Our desk review found no platform in Kumasi that lets independent Aboboyaa operators join as partners, pays them only on verified disposal, and gives the assemblies the resulting data. That combination, plus WasteCure's existing presence and customer base, is KumBola's advantage.

20. Conclusion and Request

Greater Kumasi already has the people, the vehicles and the willingness to pay needed to collect its waste properly. What it lacks is the system that connects them and proves the waste went where it should. KumBola supplies that system, using a model Kumasi residents already understand from ride-hailing, corrected by the evidence of what went wrong with it.

We request **GHS 1,725,206 (US$149,368)** over 36 months. In return the project will deliver verified collection for 20,000 households and 600 businesses, 26,195 tonnes of waste legally disposed, 127 decent livelihoods paying GHS 11,584,072 to operators, open data for the city, and a platform that pays its own way from month 27. We would welcome the opportunity to discuss the proposal, share the financial model and arrange a visit to WasteCure's operations in Kumasi.

Frank Koomson
Founder, KoomBei Digital Limited, on behalf of the KumBola partners
Dream with us. Rise with us.

Annex A: Key Assumptions and Sources

Assumption	Value	Source	Confidence
Exchange rate	GHS 11.55 per US$	Bank of Ghana via MyJoyOnline, Sept 2026	High
Blended household tariff	GHS 41.60/month	Plan mix (Section 8); Jumeni 2026 pricing benchmark	Medium
Waste per household account	0.075 t/month	0.75 kg/person/day (Miezah et al., 2015) x 3.6 persons (Ghana Statistical Service, 2021)	Medium
Household accounts, end Years 1 to 3	2,500 / 9,000 / 20,000	Project targets; 2.4% of Greater Kumasi households by month 36	Medium
Monthly customer churn	3% households, 2% businesses	Assumption	Medium-low
Commission	15%	Below Uber 25%, Bolt 20% (Akorsu, 2023)	High (policy)
Pickups per driver per day	30	Kumasi tricycles serve 46 households/day (Ampong et al., 2024)	Medium
New 200cc tricycle	GHS 30,000	Yen, 2026 (GHS 25,000 to 32,000)	High
Petrol	GHS 16.77/litre	Pulse Ghana, 16 Sept 2026	High
KMA fee and tolls	GHS 30/day + GHS 5/visit	Adom Online, 2026b	Medium
Payment processing	1.95%; GHS 1 per payout	Paystack Ghana, 2026	High
Saleable recyclables share	3% / 5% / 6% of tonnage	Plastics about 14% of Ghana's municipal waste (Miezah et al., 2015); conservative	Medium-low
Recyclables net margin	GHS 0.50/kg (sale GHS 1.20 less bonus and handling)	Al Jazeera, 2025; Graphic Online, 2026b; estimate	Low: to validate
Acquisition cost	GHS 25 per household, GHS 100 per business	Assumption	Low: to validate
Staff costs	GHS 22,500 to 46,000/month	Kumasi salary levels (assumption)	Medium
Emissions avoided	0.5 tCO2e per tonne not burned or dumped; 20% counterfactual	IPCC 2006 factors (Intergovernmental Panel on Climate Change, 2006); KMA Composite Budget 2026	Medium-low

Annex B: Assumptions to Validate During the 6-Month Pre-Enrolment Research

We have been explicit about which numbers are measured and which are estimated. The following will be confirmed with field data during the six months of research before the project is enrolled, and the financial model updated and re-shared with funders:

1. Number of active Aboboyaa waste operators in the pilot municipalities, their daily trips, fees and incomes (operator census and survey).
2. Current household fees paid to tricycle collectors in Suame, Kwabre East and Oforikrom (household survey).
3. Recyclables prices from Suame and Kaase aggregators and processors, and signed offtake terms.
4. WasteCure's verified fleet size, customer numbers and tonnages.
5. Quotes for USSD, third-party insurance for commercial tricycles, and rider licensing costs under L.I. 2519, including its provisions for cargo tricycles.
6. Approved disposal and transfer sites for each pilot assembly, given KMA's stated intention to restrict other assemblies' use of Oti.

Annex C: Funding Pathway

Funder or partner	Fit with KumBola	Typical size	Status (September 2026)
Global Innovation Fund	Pilot grants for testable, scalable innovations	Up to US$230,000	Rolling applications
develoPPP Ventures (BMZ, via DEG Impulse and GIZ)	Matching funds for young companies in Ghana	Up to €100,000	2026 programme open to Ghana
KfW Investing for Employment	Capital investment and jobs, scored on cost per job; WasteCure as lead	Varies	2026 call closed 1 June; prepare for next round
GEF Small Grants Programme Ghana (UNDP)	Chemicals and waste window for NGOs and CBOs	US$30,000 to 50,000	Next call to be announced
Environment360 / GreenCollar and plastic credit buyers	Plastic credit revenue partnership; planned Kumasi expansion	Revenue, not grant	Partnership discussion
Alliance to End Plastic Waste; GRIPE and brand owners	Recovery infrastructure and community buyback	Project-based	No open call; partnership approach
Tony Elumelu Foundation	Seed capital and training for the joint venture	US$5,000	Next window usually opens in January
Kumasi Metropolitan Assembly and assemblies	Service contracts and dashboard fees	Contract-based	MoU discussions in inception

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